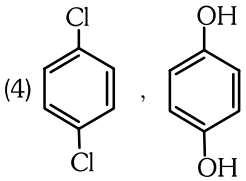


CHEMICAL BONDING

1. Which of the following can be extracted with ether?
 - (1) NaCl (2) LiCl
 - (3) KCl (4) None
2. Select the pairs for which dipole moment of first species is greater than second—
 - (1) NH_3 , NF_3
 - (2) BF_3 , BCl_3
 - (3) trans-but-2-ene, cis-but-2-ene
 - (4) 
3. Dipole moment of a diatomic molecule is 1.2 D. If its bond length is 1×10^{-8} cm, what fraction of charge exists on each atom—
 - (1) 0.1 (2) 0.2 (3) 0.25 (4) 0.3
4. Which one of the following species does not exist?
 - (1) He_2^{+2} (2) He_2 (3) He_2^+ (4) Li_2^+
5. The species having bond order different from that in 'CO' is—
 - (1) NO^- (2) NO^+
 - (3) CN^- (4) N_2
6. The bond order in NO is 2.5 while that in NO^+ is '3'. Which statement is true—
 - (1) bond length comparison is Unpredictable
 - (2) bond length in NO is greater than is NO^+
 - (3) bond length in NO^+ is equal to that in NO
 - (4) bond length in NO^+ is greater than in NO
7. The maximum No. of 90° angles between bp-bp of electrons is observed in—
 - (1) sp^3d^2 Hybridisation (2) sp^3d Hybridisation
 - (3) dsp^2 Hybridisation (4) dsp^3 Hybridisation
8. The shape of ' AsF_5 ' according to V.S.E.P.R theory—
 - (1) Pentagonal Pyramidal
 - (2) Trigonal bipyramidal
 - (3) Square Pyramidal
 - (4) Octahedral
9. CO_2 is linear but H_2O is bent, because—
 - (1) CO_2 is polar, H_2O is not
 - (2) H_2O is polar, CO_2 is not
 - (3) Carbon has low E.N than 'H'
 - (4) Carbon can form double bond with 'O'
10. If 'X'-axis is internuclear axis, which of the following orbitals will not form a ' σ ' bond—
 - (1) 1s & 1s (2) 1s & $2p_x$
 - (3) $2p_y$ and $2p_y$ (4) 1s and 2s
11. The Hybrid state of 'starred' carbon (*)

$\text{CH}_3 \overset{*}{\text{C}} \text{HO}$

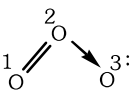
 - (1) sp (2) sp^2 (3) sp^3 (4) sp^3d
12. Condition for Linear combination of Atomic orbitals according to M.O.T is—
 - (1) Orbital must have same symmetry
 - (2) Orbitals must have same energy
 - (3) Orbitals must overlap to maximum extent
 - (4) All
13. In PCl_5 , axial bonds are longer than equatorial bonds because—
 - (1) Equatorial bonds lie in same plane
 - (2) Axial bonds are repelled more by equatorial bonds
 - (3) Irregular electron distribution in 'P'
 - (4) There is bond angle of 120° between equatorial bonds.
14. Which lewis dot structure for O^{2-} ion is correct ?

$(1) \begin{array}{c} \cdot\ddot{\text{O}}\cdot \\ \cdot\ddot{\text{O}}\cdot \end{array}$

$(2) \left[\begin{array}{c} \cdot\ddot{\text{O}}\cdot \\ \cdot\ddot{\text{O}}\cdot \end{array} \right]^{2-}$

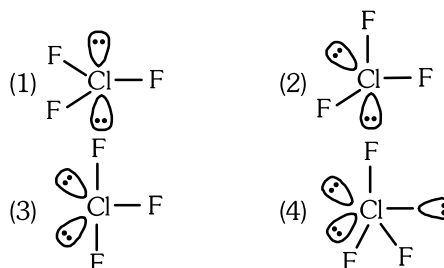
$(3) \left[\cdot\ddot{\text{O}}\cdot \right]^{2-}$

$(4) \left[\begin{array}{c} \cdot\ddot{\text{O}}\cdot \\ \cdot\ddot{\text{O}}\cdot \end{array} \right]^{2-}$
15. Sodium and Chlorine atoms combines—
 - (1) to acquire electronic configuration of argon.
 - (2) to acquire highest oxidation states.
 - (3) by exchanging valence electrons.
 - (4) to complete their octets.

- 16** The group valency of the elements generally equals to—
 (1) Total number of electrons
 (2) 8 minus, total numbers of electrons
 (3) Number of valence electrons or 8 minus valence electrons
 (4) higher oxidation state of the elements
- 17** In ammonium ion, bond in between ammonia molecule and a proton is formed by—
 (1) Complete transfer of electron from NH_3 to H^+
 (2) electrostatic attraction between NH_4^+ & H^+
 (3) equal contribution of electrons by NH_3 & H^+
 (4) One sided sharing of electrons
- 18** The correct structure of CO and NO_2^- are—
 (1) $\text{:C}\equiv\ddot{\text{O}}\text{:}, \ddot{\text{O}}=\ddot{\text{N}}=\ddot{\text{O}}$
 (2) $\text{:C}\equiv\text{O}\text{:}, \left[\ddot{\text{O}}=\ddot{\text{N}}-\ddot{\text{O}}\text{:} \right]^-$
 (3) $\text{:C}\equiv\ddot{\text{O}}\text{:}, \left[\text{:O}=\ddot{\text{N}}\rightarrow\ddot{\text{O}}\text{:} \right]^-$
 (4) $\text{:}\ddot{\text{C}}\equiv\ddot{\text{O}}\text{:}, \left[\text{:O}=\text{O}\text{:}\rightarrow\text{N} \right]^-$
- 19** Lewis structure of O_3 is drawn as  therefore formal charge on oxygen atoms are—
 (1) 0, 0, 0 (2) 0, +1, -1
 (3) 0, +1, +1 (4) -1, +1, -1
- 20** Which among the following is diamagnetic molecule and follow octet rule?
 (1) NO_2 (2) ClO_2 (3) CO_2 (4) NO
- 21** Elements in and beyond the third period can expand their octet in the molecules like PF_5 , SF_6 etc, because
 (1) they contain more than eight-electron in the outermost shell.
 (2) their atomic size is large.
 (3) they have vacant $(n-1)d$ orbitals.
 (4) They have vacant nd orbitals.
- 22** The ionic bond is formed by—
 (1) Complete transfer of electrons from cation to anion
 (2) Exchanging electrons in between metal and nonmetal
 (3) donation of electron pair from electro positive to electronegative element.
 (4) Complete transfer of 1, 2 or 3 electron from electropositive to electronegative element.
- 23** The Lewis theory does not account for the—
 (1) cause of bond formation
 (2) Shape of molecules
 (3) Strength of chemical bond
 (4) All
- 24** Stability of ionic crystal is mainly affected by—
 (1) Lattice energy (2) ΔEN
 (3) IP (4) EA
- 25** Bond enthalpy in H_2 molecule is $435.8 \text{ KJ mole}^{-1}$ where as bond enthalpy in N_2 molecule is $946.0 \text{ KJ mole}^{-1}$, because
 (1) H_2 contains ionic bond where as N_2 covalent.
 (2) H_2 does not have lone pair but N_2 have.
 (3) Internuclear distance in H_2 is larger than N_2 .
 (4) H_2 contains single bond where as N_2 contains multiple bond.
- 26** Isoelectronic molecules and molecular ions have—
 (1) Same bond energy
 (2) identical bond order
 (3) same charge on molecules
 (4) same bond length
- 27** Which among the following statements is wrong?
 (1) Resonance stabilizes the molecules
 (2) Energy of resonance hybrid is more than the single canonical structure
 (3) Resonance averages the bond characteristics as a whole.
 (4) The canonical form have no real existence.
- 28** In polyatomic molecules dipole moment depend upon—
 (1) Individual dipole moment of bonds.
 (2) Spatial arrangement of various bonds.
 (3) Vector sum of the dipole moments of various bonds.
 (4) Geometry of molecules.
- 29** Which of the following statement regarding NH_3 and NF_3 is correct ?
 (1) NH_3 has pyramidal and, NF_3 has trigonal planar shape
 (2) bond angle in NH_3 is smaller than NF_3
 (3) resultant dipole moment of NH_3 is $(4.90 \times 10^{-30} \text{ cm})$ and that of NF_3 is $(0.8 \times 10^{-30} \text{ cm})$
 (4) they both are sp^2 hybridized.

- 30** The cations having same size and charge one having $(n-1)d^{10}ns^0$ configuration possess more polarising power, because is—
- (1) It is d-block element
 - (2) It has inert gas configuration
 - (3) It has effective shielding of d-electrons
 - (4) it has more Z_{eff} due to poor shielding of d-electrons.
- 31** According to VSEPR theory the pairs of electrons tend to occupy such position in space that—
- (1) Maximise repulsion and thus minimise distance from one another.
 - (2) minimise repulsion and thus maximise distance from one another.
 - (3) maximise repulsion and thus maximise distance from one another.
 - (4) minimise repulsion and distance from one another.
- 32** Which of the following molecule does not contain bond angle of 90° ?
- | | |
|--------------------|---------------------|
| (1) IF_7 | (2) SiCl_4 |
| (3) XeF_4 | (4) PCl_5 |

- 33** Which among the following structure for ClF_3 is most stable ?



- 34** In the reaction $\text{BF}_3 + \text{NH}_3 \longrightarrow \text{F}_3\text{B} \cdot \text{NH}_3$
- (1) Bond angle $\text{F} - \text{B} - \text{F}$ increase
 - (2) $\text{H}\hat{\text{N}}\text{H}$ bond angle decrease
 - (3) regular geometry of BF_3 changed
 - (4) regular geometry of NH_3 changed.
- 35** The correct increasing order of bond order is—
- (1) $\text{N}_2 < \text{O}_2 < \text{O}_2^+ < \text{O}_2^-$
 - (2) $\text{O}_2^- < \text{O}_2 < \text{O}_2^+ < \text{N}_2$
 - (3) $\text{O}_2^+ < \text{O}_2 < \text{O}_2^- < \text{N}_2$
 - (4) $\text{O}_2 < \text{N}_2 < \text{O}_2^+ < \text{O}_2^-$